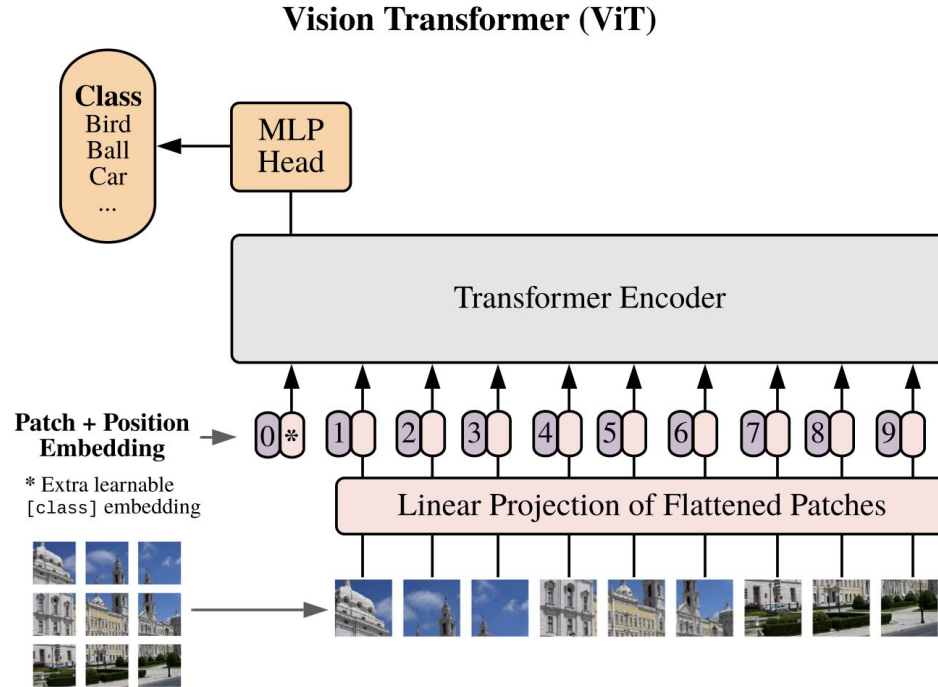


Transformers in Deep Learning

Lecture 5

Lecture 4 Recap

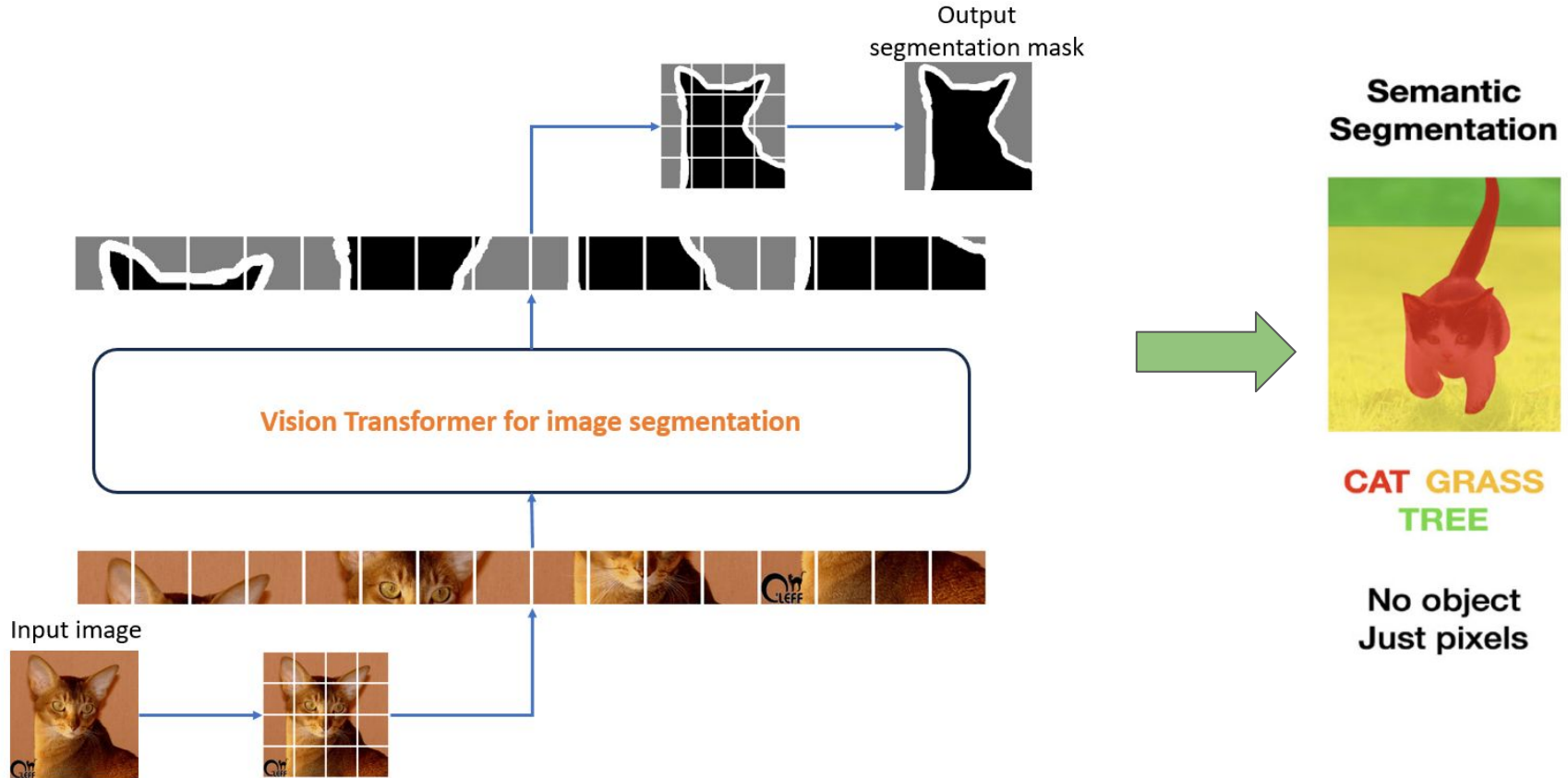


Classification

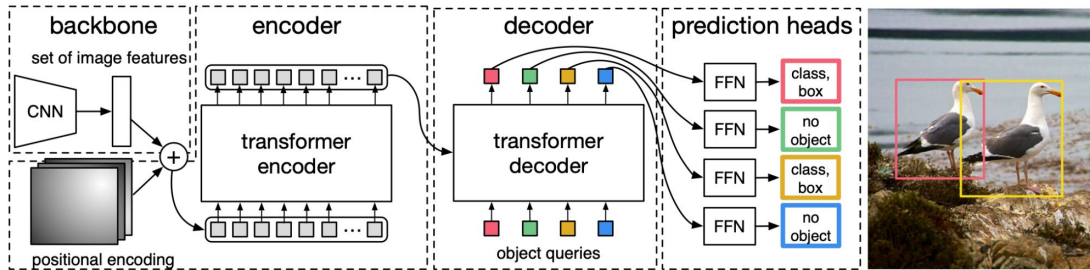


CAT

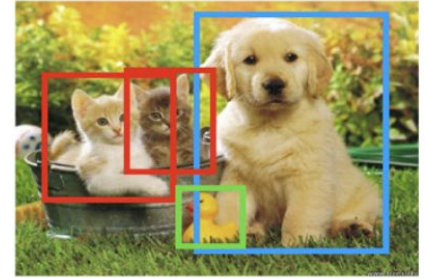
Lecture 4 Recap



Lecture 4 Recap



Object detection

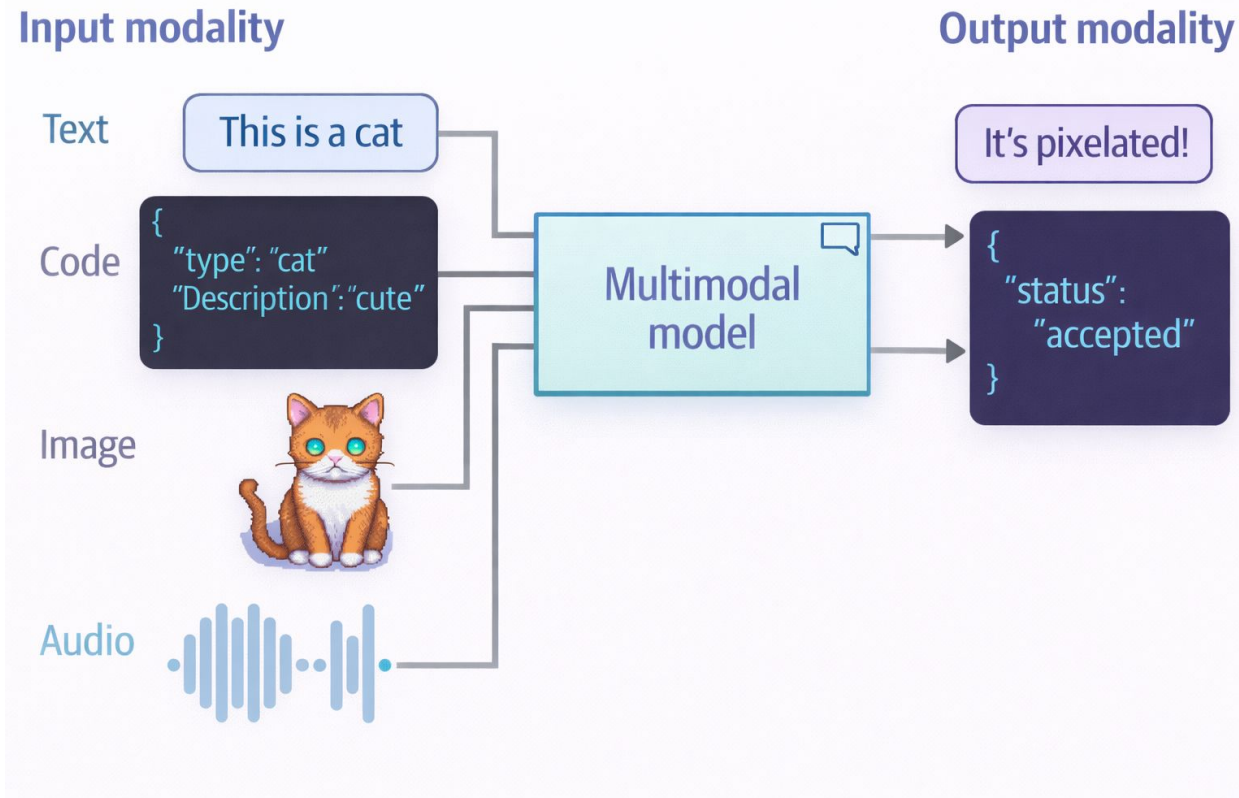


CAT DOG DUCK

One hammer for everything:



Multimodality

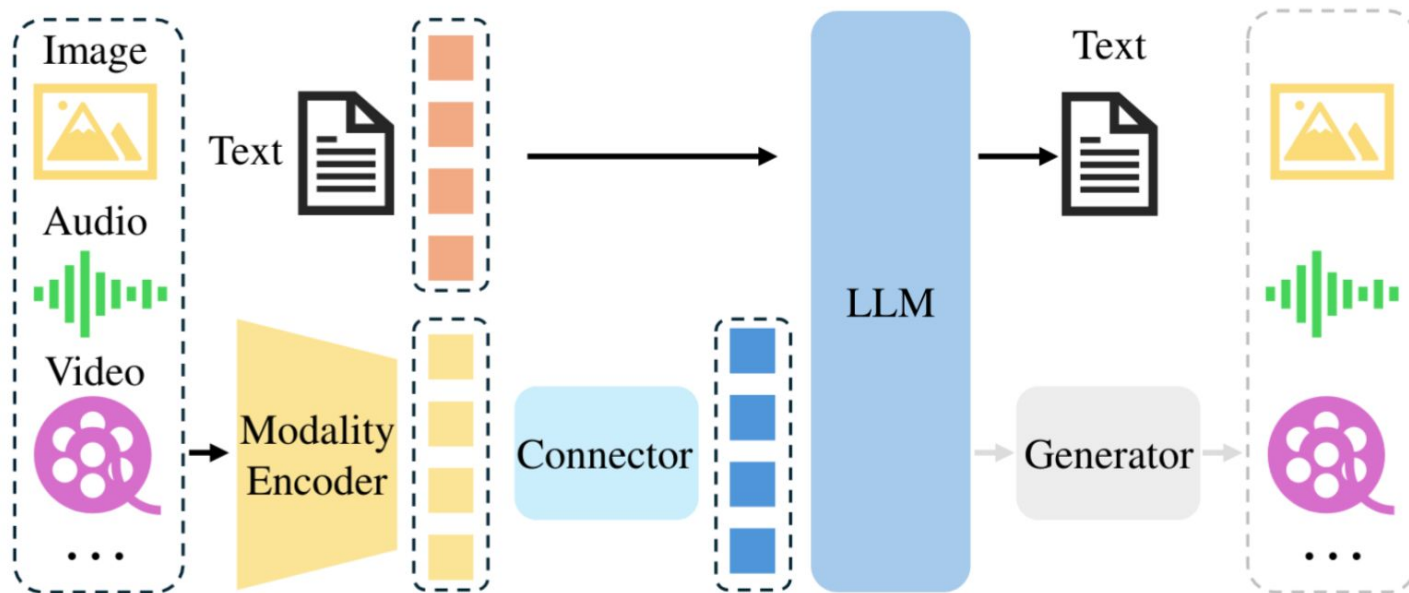


Multimodality: key ideas

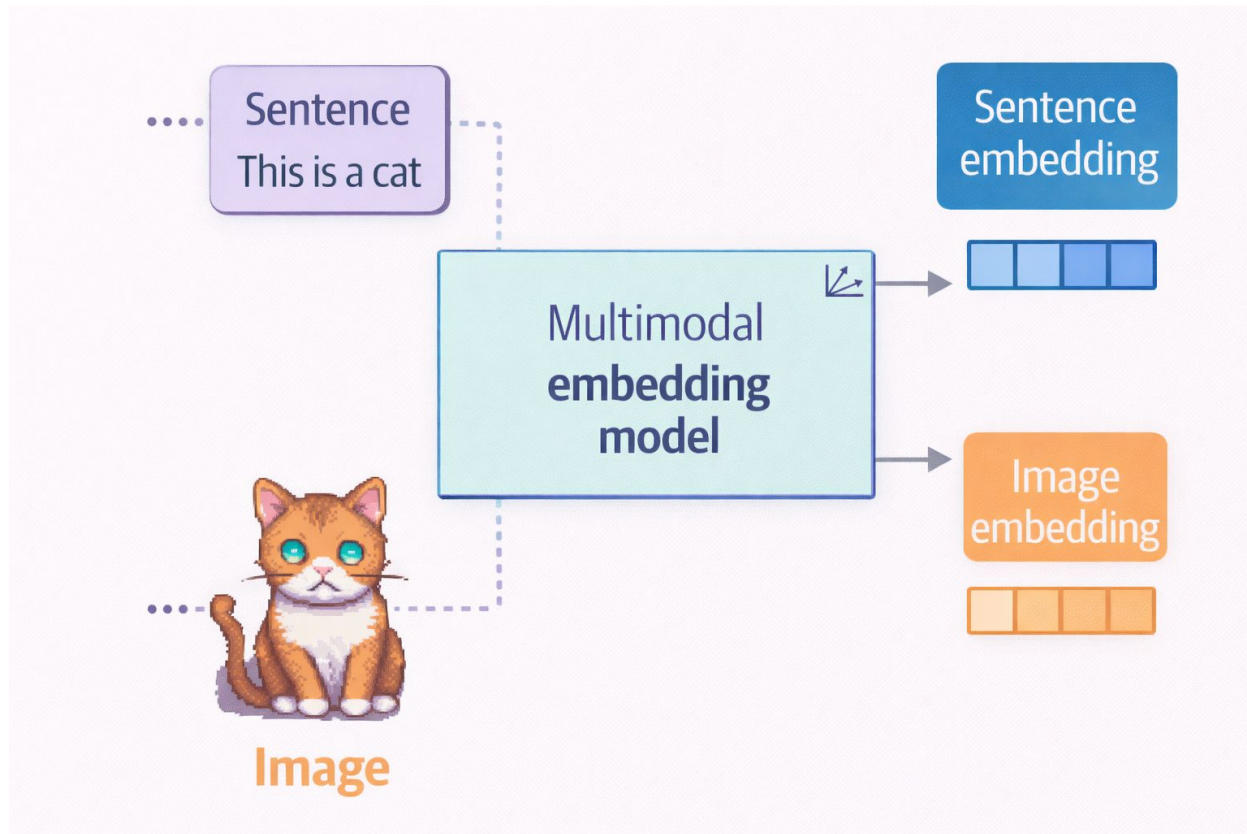
- Use world knowledge and reasoning from LLMs
- Explore various applications that emerge at the intersection of the modalities
- Boost quality of models using patterns in multimodal data that don't exist in unimodal data



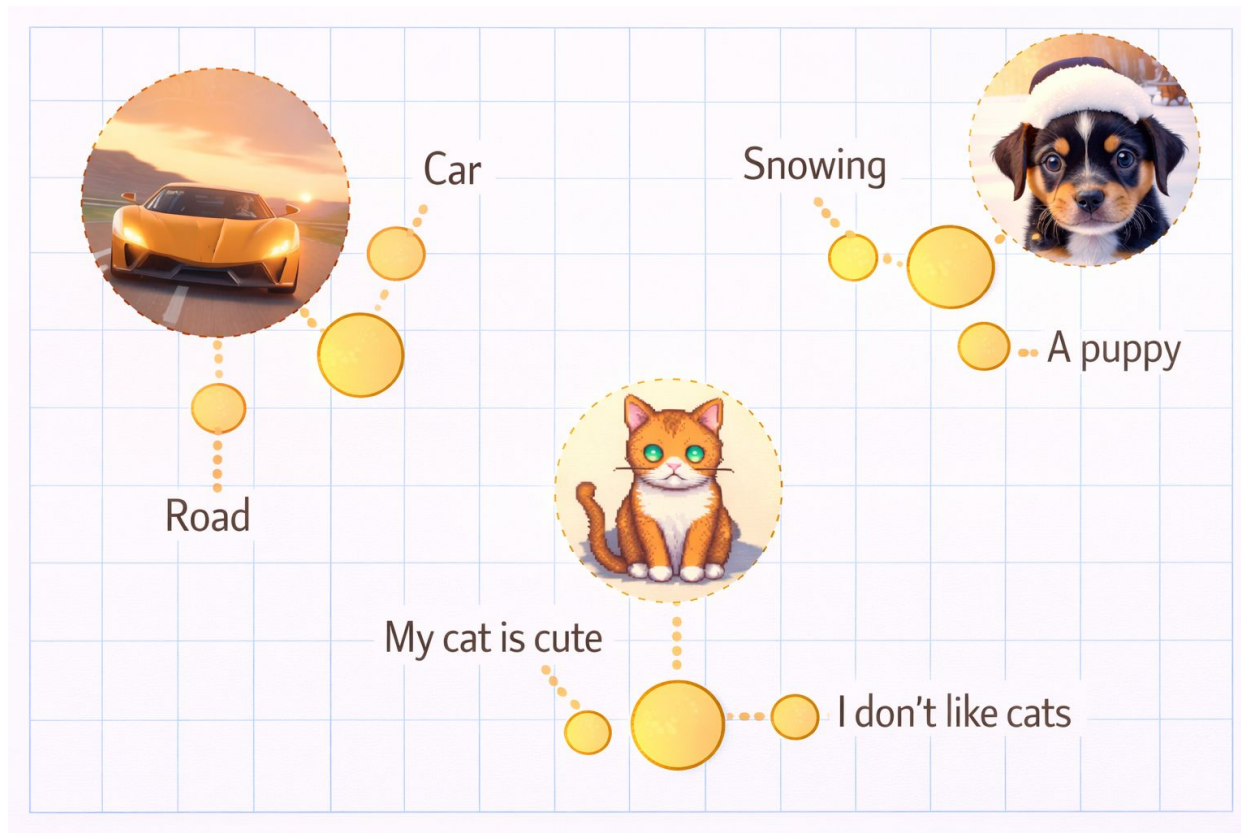
General scheme



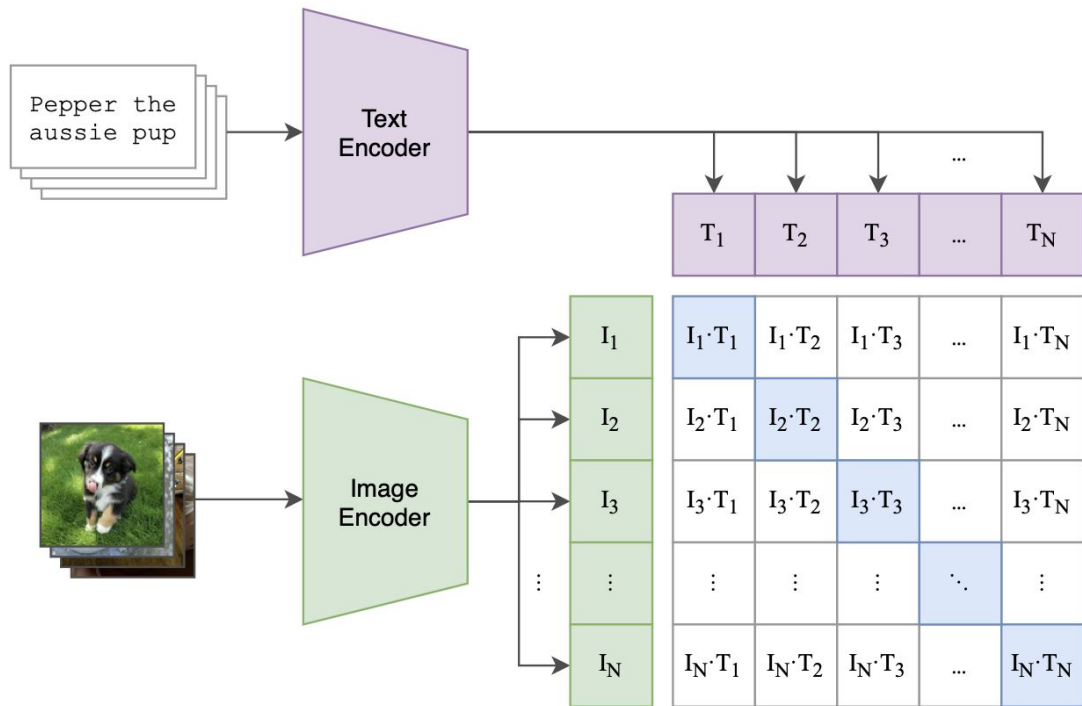
Multimodal embedding model



Multimodal embedding space



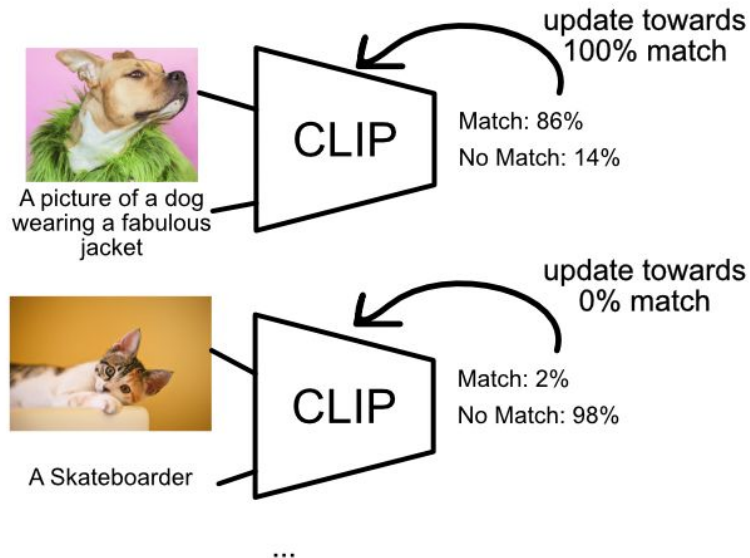
Multimodal embedding model: CLIP



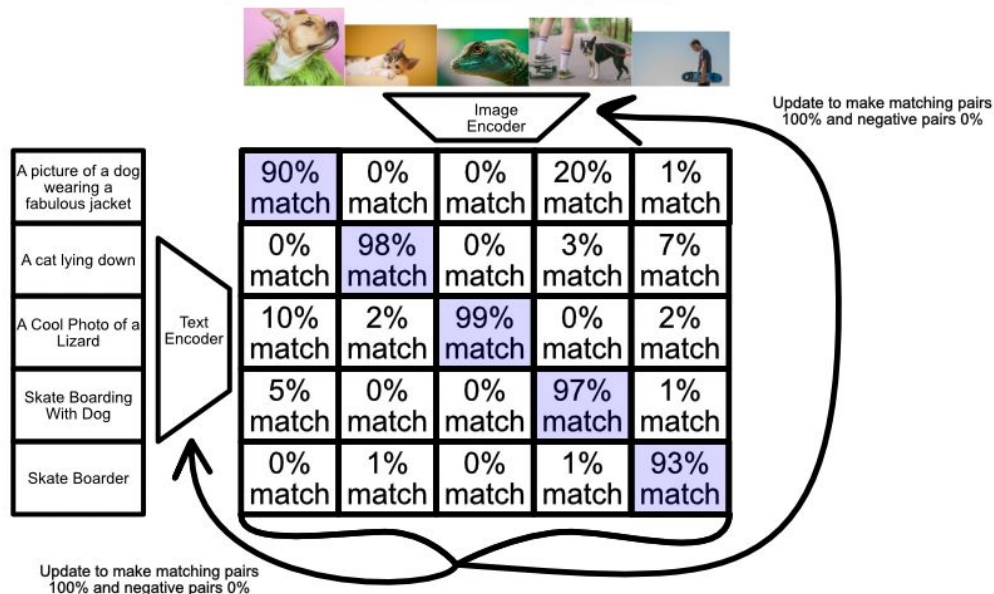
400M (image, text) pairs, 500×V100 GPUs for pretraining

CLIP: Contrastive Learning

Typical Supervised Approach



Contrastive Supervised Approach



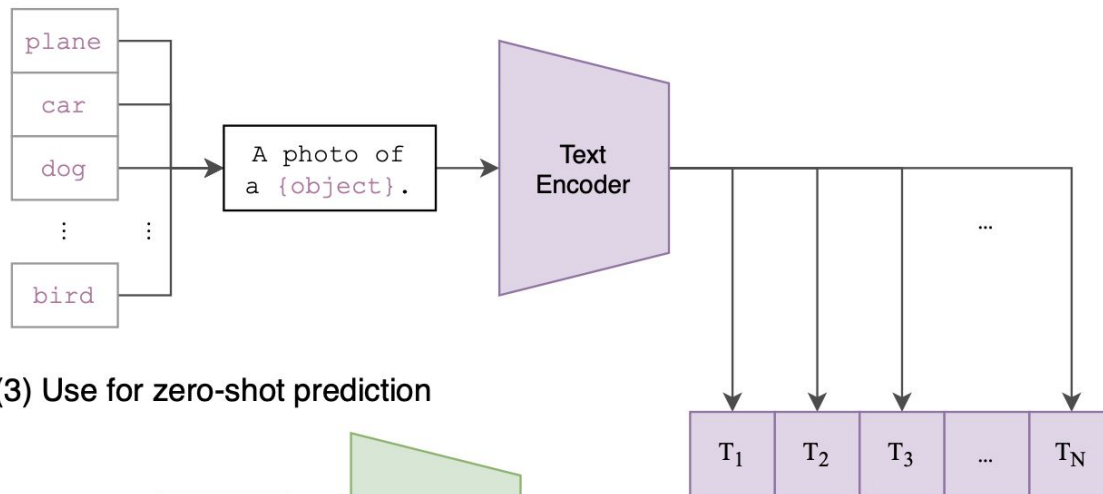
CLIP: Contrastive Loss

CLIP: Softmax

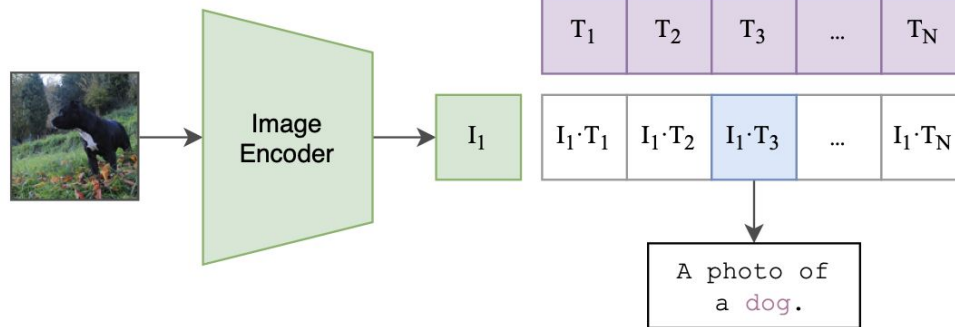
$$-\frac{1}{2N} \sum_{i=1}^N \left(\overset{\text{Image} \rightarrow \text{text softmax}}{\log \frac{e^{I_i \cdot T_i / \tau}}{\sum_{j=1}^N e^{I_i \cdot T_j / \tau}}} + \overset{\text{text} \rightarrow \text{image softmax}}{\log \frac{e^{I_i \cdot T_i / \tau}}{\sum_{j=1}^N e^{I_j \cdot T_i / \tau}}} \right)$$

CLIP

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



SigLIP: better CLIP

CLIP: Softmax

$$-\frac{1}{2N} \sum_{i=1}^N \left(\overset{\text{Image} \rightarrow \text{text softmax}}{\log \frac{e^{I_i \cdot T_i / \tau}}{\sum_{j=1}^N e^{I_i \cdot T_j / \tau}}} + \overset{\text{text} \rightarrow \text{image softmax}}{\log \frac{e^{I_i \cdot T_i / \tau}}{\sum_{j=1}^N e^{I_j \cdot T_i / \tau}}} \right)$$

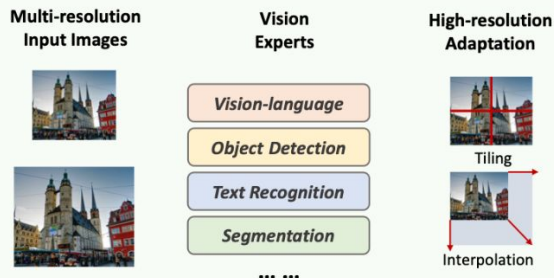
SigLIP: Sigmoid

$$-\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \log \frac{1}{1 + e^{z_{ij}(I_i \cdot T_j / \tau + b)}}$$

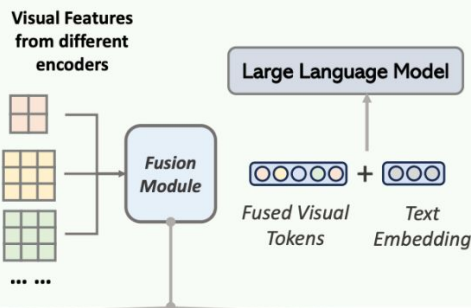
- turn multiclass classification into binary classification of all pair (image, text) combinations
- no global normalization, hence better scaling and memory efficiency

Mixture of Encoders (Visual) (MOE)

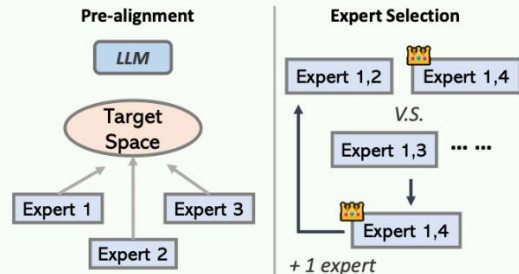
Step1: Vision Encoder Modification Optimization



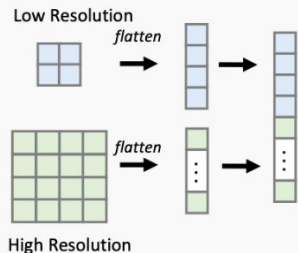
Step2: Fusion Paradigm Exploration



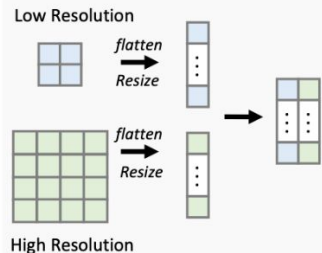
Step3: Training Strategy and Model Optimization



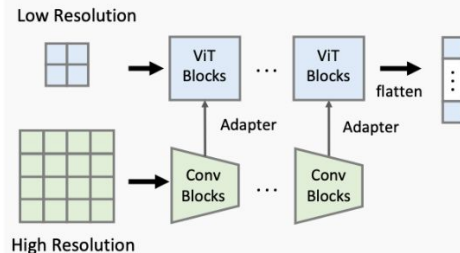
Sequence Append



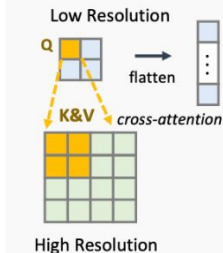
Channel-wise Concatenation



LLAVA-HR



Mini-Gemini



Deformable Attention



High-level Classification

Multimodal models can be classified in 2 main types (4 subtypes)
based on the **fusion of input modalities**

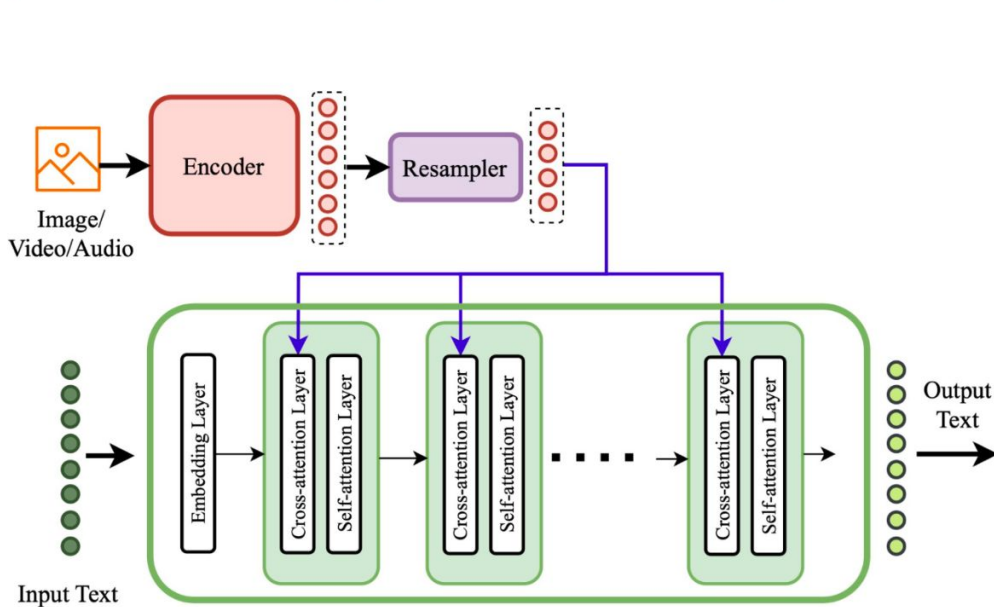


Deep Fusion:

Standard Cross-Attention
Deep Fusion (SC-DF)

SC-DF: Standard Cross-Attention

Input modalities are deeply fused into the **internal layers of the LLM** using **standard cross-attention layer**



before

Flamingo

OpenFlamingo

Otter

Multimodal-GPT



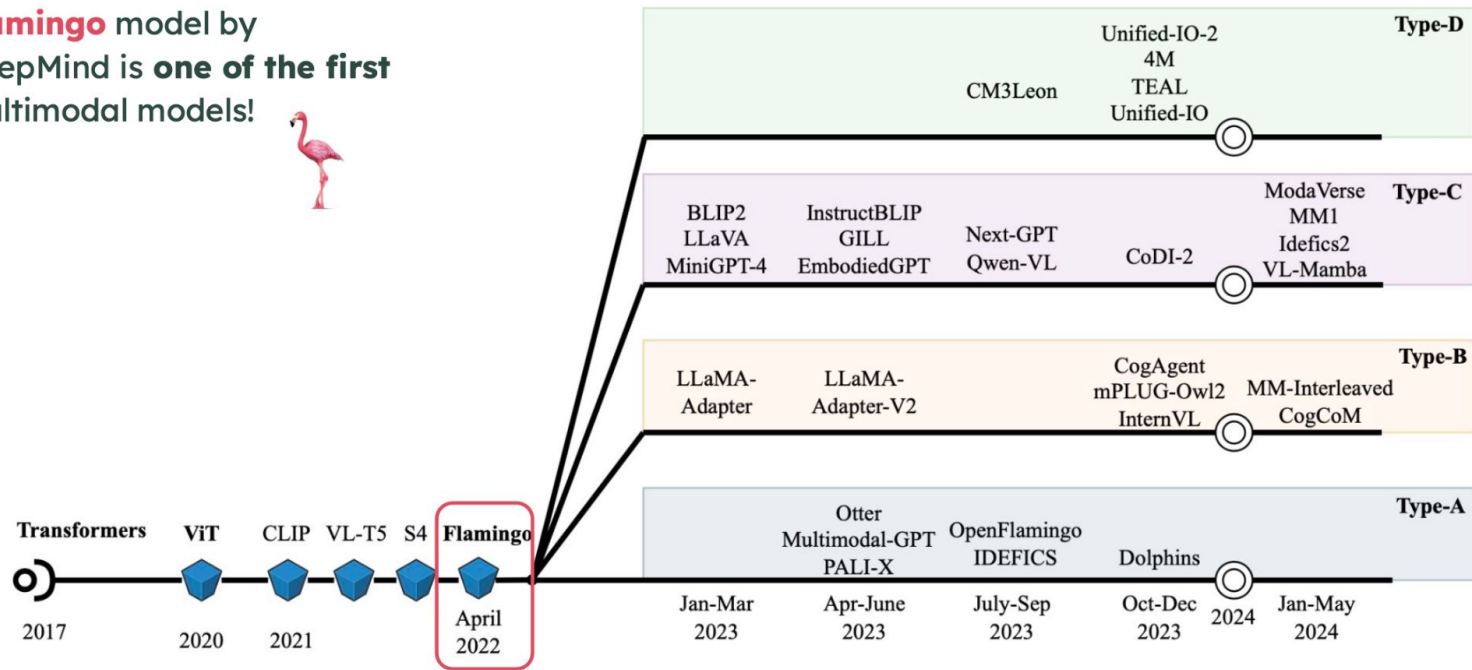
after

VL-BART

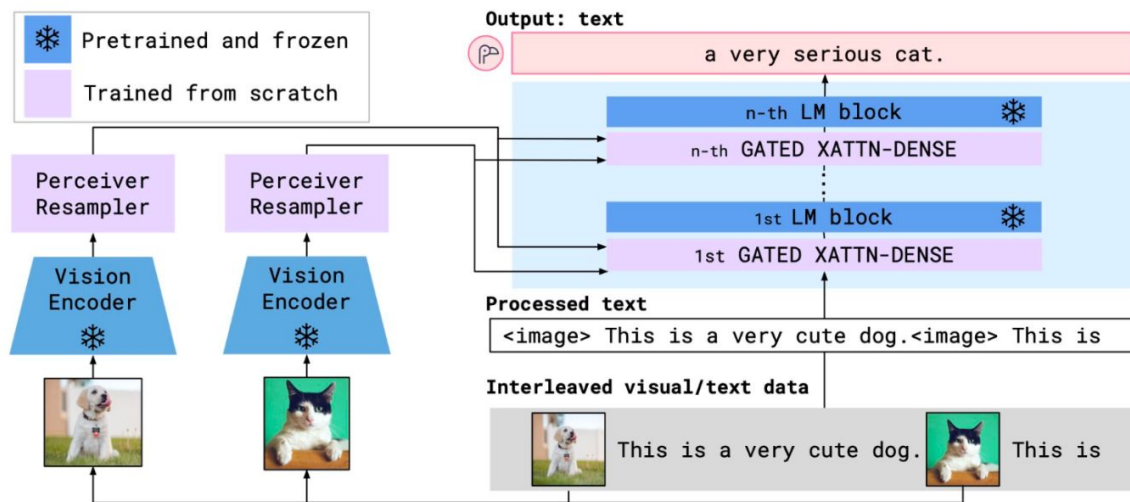
VL-T5

SC-DF: Standard Cross-Attention

Flamingo model by DeepMind is **one of the first** multimodal models!



SC-DF: Open-Flamingo (nov 2022)



vision model
CLIP ViT-L/14
(NFNet)

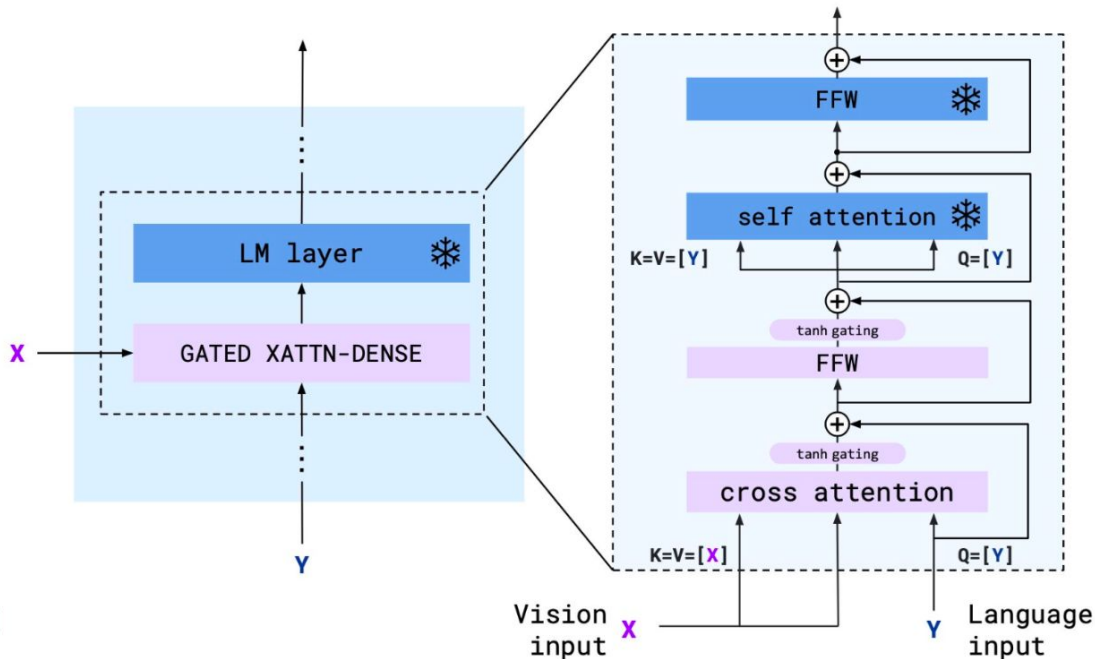
language model
RedPajama / MPT
(Chinchilla)

OpenFlamingo follows Flamingo architecture

SC-DF: Open-Flamingo (nov 2022)

1. **freeze** the pretrained LM blocks
2. insert **gated cross-attention dense** blocks between the original layers
3. keep layers gated to keep LM intact at initialization
4. **queries** = LM inputs

tanh-gating mechanism —
multiplies output of newly initialized
layer by $\tanh(\alpha)$

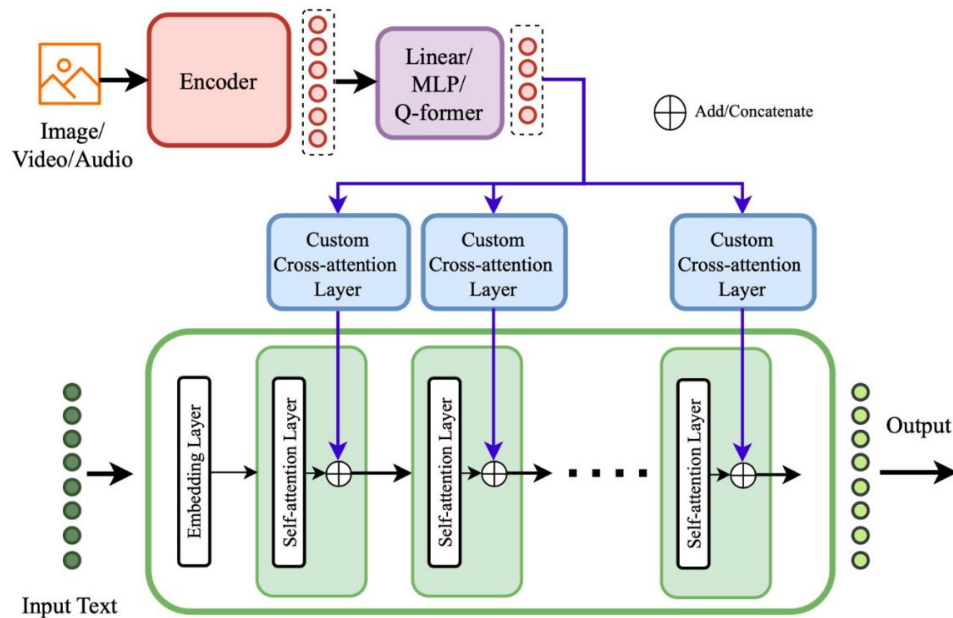


Deep Fusion:

Custom Layers Deep Fusion
(CL-DF)

CL-DF: Custom Layers

Input modalities are deeply fused into the **internal layers of the LLM** using **custom-designed** layers



custom
cross-attention

LLaMA-Adapter
CogVLM
InternVL
mPLUG-Owl2

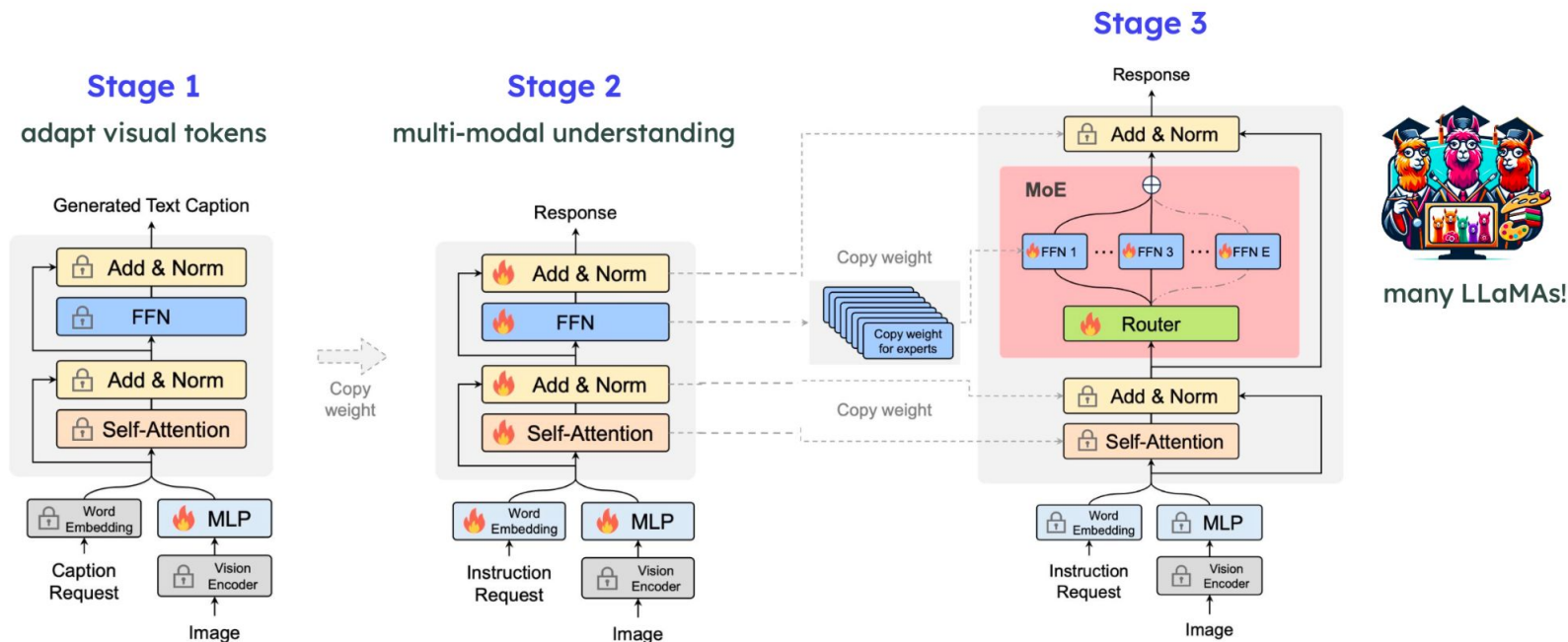
other custom
layers

MoE-LLaVA
LION

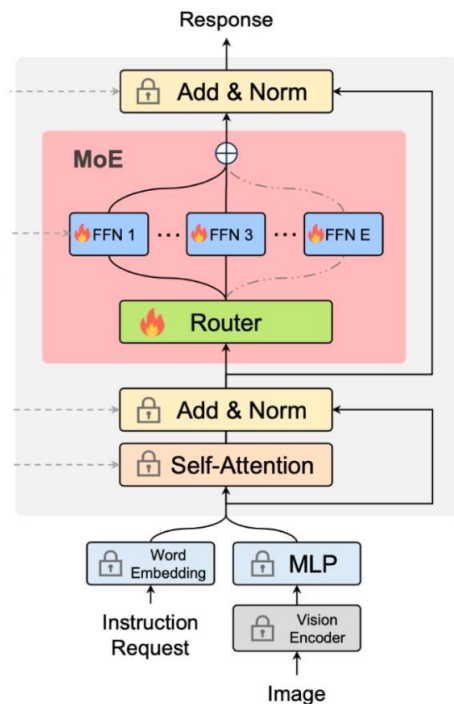


CL-DF: MoE-LLaVA (dec 2024)

LLaVA – by Microsoft, **MoE-LLaVA** – Peking University: Mixture-of-Experts layer



MoE-LLaVA: Router



1

have E experts, each expert = FFN

$$\mathcal{E} = [e_1, e_2, \dots, e_E]$$

2

router = linear layer that assigns probabilities to experts

$$\mathcal{P}(\mathbf{x})_i = \frac{e^{f(\mathbf{x})_i}}{\sum_j^E e^{f(\mathbf{x})_j}}$$

3

calculate weighted sum

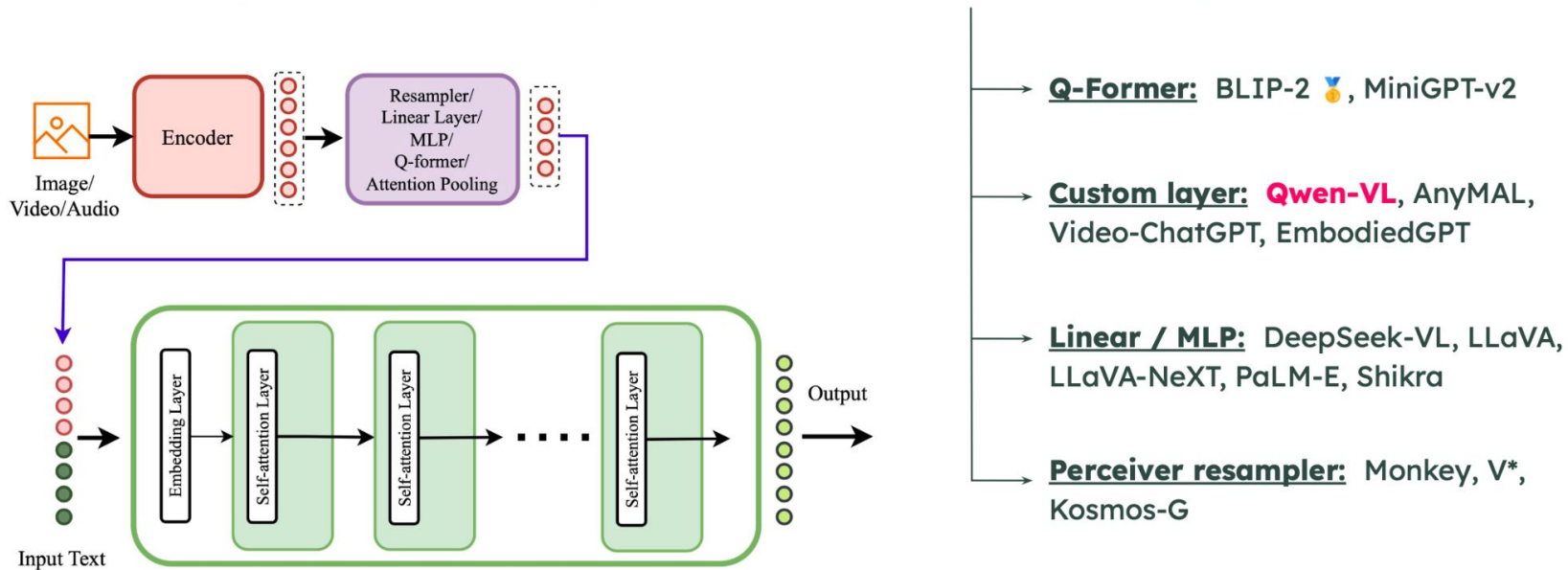
$$\text{MoE}(\mathbf{x}) = \sum_{i=1}^k \mathcal{P}(\mathbf{x})_i \cdot \mathcal{E}(\mathbf{x})_i$$

Early Fusion:

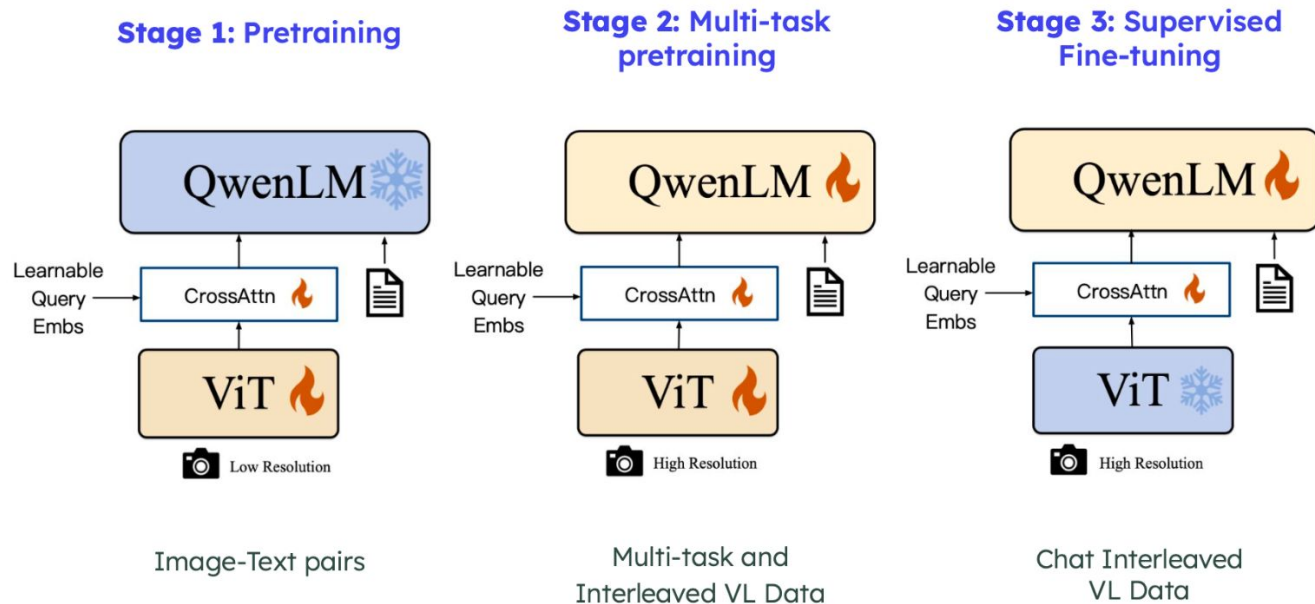
Non-Tokenized Early Fusion
(NT-EF)

NT-EF: Non-Tokenized

Non-tokenized input modalities are **directly fed to the model** rather than to internal layers



NT-EF: Qwen-VL (oct 2023)



vision model
OpenClip ViT-bigG

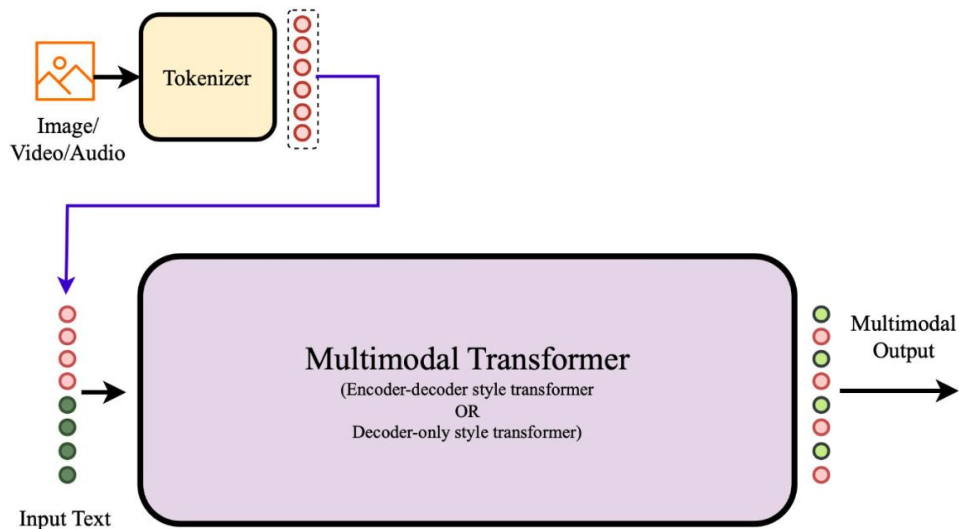
language model
Qwen-7B

Early Fusion:

Tokenized Early Fusion (T-EF)

T-EF: Tokenized

Inputs are tokenized **using a common tokenizer** or modality specific tokenizers



decoder-only
transformer

LaVIT
TEAL
CM3Leon
VL-GPT

encoder-decoder
transformer

Unified-IO
Unified-IO
4M

T-EF: LaVIT (mar 2024)

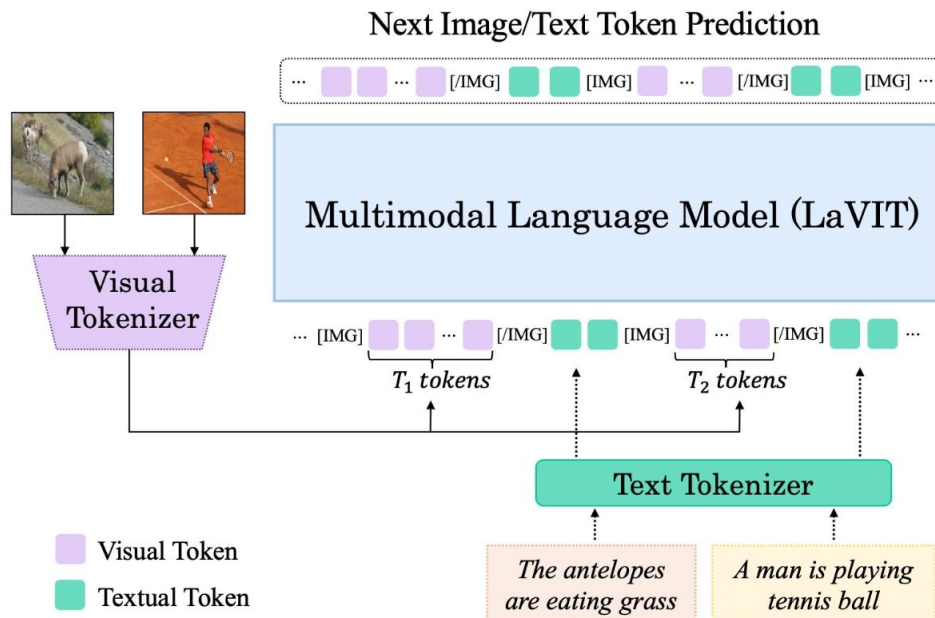
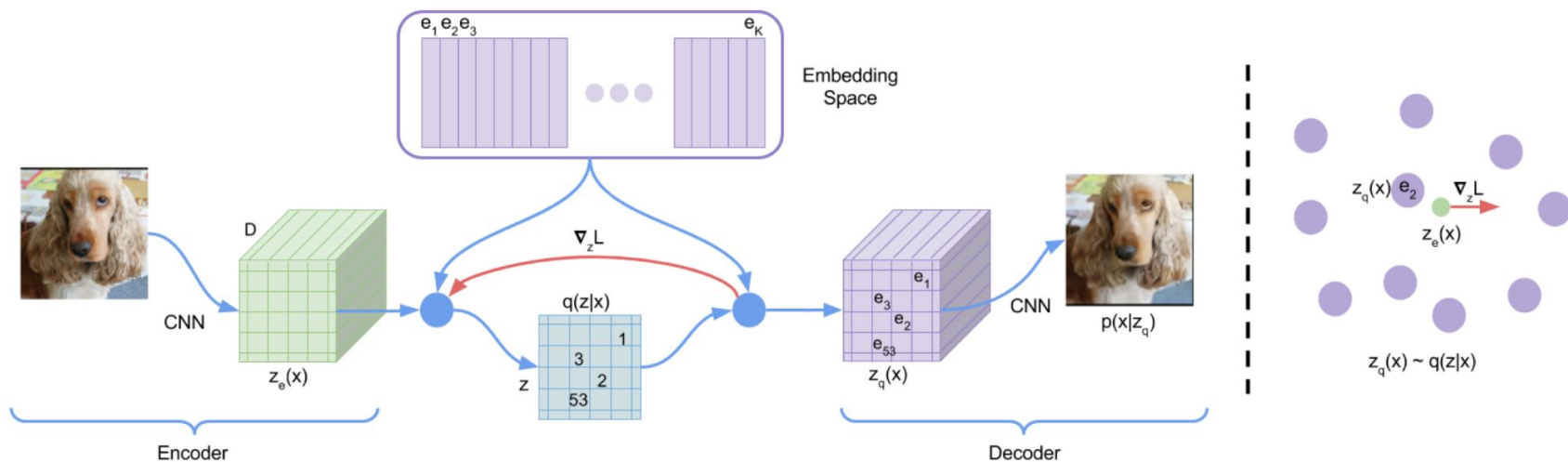


Image tokenization: VQ-VAE



Benchmarks

Static benchmarks: GQA



Pattern: What/Which <type> [do you think] <is> <dobject>, <attr> or <decoy>?

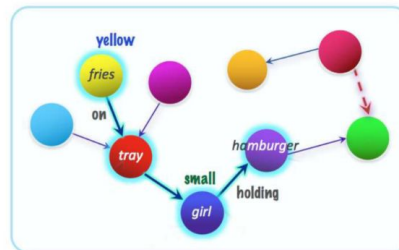
Program: Select: <dobject> → Choose <type>: <attr> | <decoy>

Reference: The food on the red object left of the small girl that is holding a hamburger

Decoy: brown

What color is the food on the red object left of the small girl that is holding a hamburger, yellow or brown?

Select: hamburger → Relate: girl, holding → Filter size: small → Relate: object, left → Filter color: red → Relate: food, on → Choose color: yellow | brown



Graph Normalization

- Ontology construction
- Edge Pruning
- Object Augmentation
- Global Properties

Question Generation

- Pattern Collection
- Compositional References
- Decoy Selection
- Probabilistic Generation

Sampling and Balancing

- Distribution Balancing
- Type-Based Sampling
- Deduplication

Entailment Relations

- Functional Programs
- Entailment Relations
- Recursive Reachability

New Metrics

- Consistency
- Validity & Plausibility
- Distribution
- Grounding

Hudson et al. GQA: A New Dataset for Real-World Visual Reasoning and Compositional Question Answering. CVPR 2019

Benchmarks

Static benchmarks: GQA



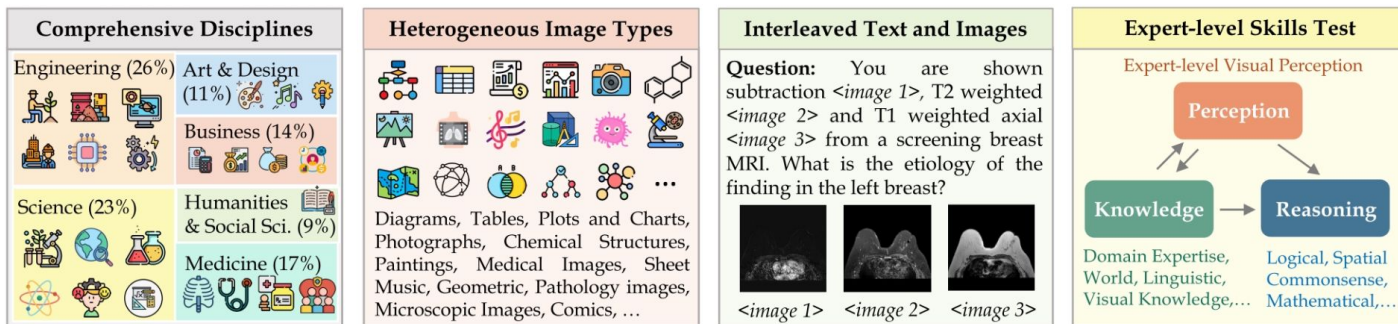
- A1. Is the **tray** on top of the **table** black or light brown? light brown
A2. Are the **napkin** and the **cup** the same color? yes
A3. Is the small **table** both oval and wooden? yes
A4. Is there any **fruit** to the left of the **tray** the **cup** is on top of? yes
A5. Are there any **cups** to the left of the **tray** on top of the **table**? no
B1. What is the brown **animal** sitting inside of? **box**
B2. What is the large **container** made of? cardboard
B3. What **animal** is in the **box**? **bear**
B4. Is there a **bag** to the right of the green **door**? no
B5. Is there a **box** inside the plastic **bag**? no

- questions generated using scene graph of images
- 22.6M questions for 113k images
- evaluation metric: accuracy along with 5 more detailed metrics

Benchmarks

Static benchmarks: MMMU

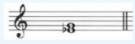
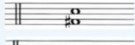
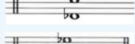
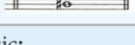
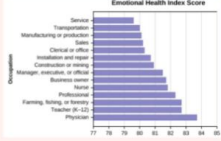
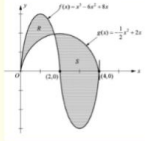
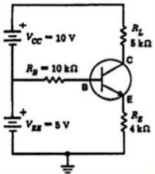
- 11.5k questions from 6 university disciplines
- answers are extracted using regexps
- evaluation metric: accuracy



Yue et al. MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI. CVPR 2024

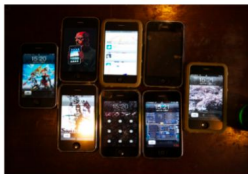
Benchmarks

Static benchmarks: MMMU

Art & Design	Business	Science
Question: Among the following harmonic intervals, which one is constructed incorrectly? Options: (A) Major third  (B) Diminished fifth  (C) Minor seventh  (D) Diminished sixth 	Question: ...The graph shown is compiled from data collected by Gallup  Options: (A) 0 (B) 0.2142 (C) 0.3571 (D) 0.5	Question:  The region bounded by the graph as shown above. Choose an integral expression that can be used to find the area of R. Options: (A) $\int_0^{1.5} [f(x) - g(x)] dx$ (B) $\int_0^{1.5} [g(x) - f(x)] dx$ (C) $\int_0^2 [f(x) - g(x)] dx$ (D) $\int_0^2 [g(x) - x(x)] dx$
Subject: Music; Subfield: Music; Image Type: Sheet Music; Difficulty: Medium	Subject: Marketing; Subfield: Market Research; Image Type: Plots and Charts; Difficulty: Medium	Subject: Math; Subfield: Calculus; Image Type: Mathematical Notations; Difficulty: Easy
Health & Medicine	Humanities & Social Science	Tech & Engineering
Question: You are shown subtraction , T2 weighted  and T1 weighted axial  from a screening breast MRI. What is the etiology of the finding in the left breast? Options: (A) Susceptibility artifact (B) Hematoma (C) Fat necrosis (D) Silicone granuloma	Question: In the political cartoon, the United States is seen as fulfilling which of the following roles?  Option: (A) Oppressor (B) Imperialist (C) Savior (D) Isolationist	Question: Find the VCE for the circuit shown in . Answer: 3.75 Explanation: ...$IE = [(V_{EE}) / (R_E)] = [(5 \text{ V}) / (4 \text{ k-ohm})] = 1.25 \text{ mA}$; $V_{CE} = V_{CC} - I_{ERL} = 10 \text{ V} - (1.25 \text{ mA}) 5 \text{ k-ohm}$; $V_{CE} = 10 \text{ V} - 6.25 \text{ V} = 3.75 \text{ V}$
Subject: Clinical Medicine; Subfield: Clinical Radiology; Image Type: Body Scans: MRI, CT,; Difficulty: Hard	Subject: History; Subfield: Modern History; Image Type: Comics and Cartoons; Difficulty: Easy	Subject: Electronics; Subfield: Analog electronics; Image Type: Diagrams; Difficulty: Hard

Benchmarks

Static benchmarks: TextVQA

			
What does it say near the star on the tail of the plane?		What is the time on bottom middle phone?	
Ground Truth	Prediction	Ground Truth	Prediction
jet	nothing	15:20	12:00
			
What is the top oz?		What is the largest denomination on table?	
Ground Truth	Prediction	Ground Truth	Prediction
16	red	500	unknown

- 45k questions for 28k images, 10 answers per question
- evaluation metric: VQA accuracy (100% correct if 3 humans provided that answer)

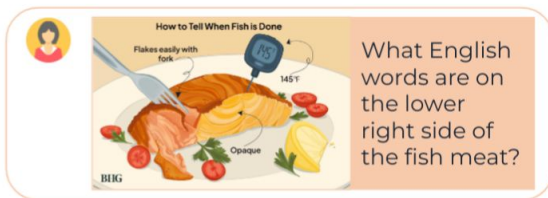
Singh et al. Towards VQA Models That Can Read. CVPR 2019

Benchmarks

WildVision-Arena

Rules

- Chat with two anonymous models
- Continue to chat until you identify a winner
- Vote for the better one with reason



Model A

On the lower right side of the cooked fish, the word "Opaque" is labeled.

Model B

The English word on the lower right side of the fish meat is "Opaque."

Reason

Both Model A and Model B answer correctly regarding the text.

Vote



A is Better



B is Better



Tie



Both are bad

Model A: Claude-3-Sonnet, Model B: GPT-4V

WVArena Elo Ratings



Submit

WVBench Scores



Judge
GPT-4o



Both models
are correct



Reference

Claude-3-Sonnet



Bench Data

500 sample



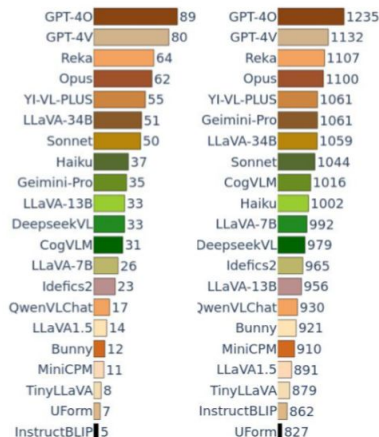
Sample Criteria

- Safety
- Diversity



Arena Data

20k+ chat 8k+ vote



WildVision
Bench

WildVision
Arena

WVArena 0.94 0.86 0.79 0.79 0.77

WVBench MMVet MMMU MMStar A2D
Correlation w. WVArena Leaderboard

Benchmarks

Statistic	Number
Total Votes	8,076
Anonymous	6,636
Non-anonymous	1,440
Left Vote	2,932
Right Vote	2,839
Tie Vote	979
Bad Vote	1,326
Days	102
Total Round	10,884
Avg Round	1.34
Avg Token Input	31.00
Avg Token Output	108.87

Table 1: Statistics of votings in WV-ARENA.

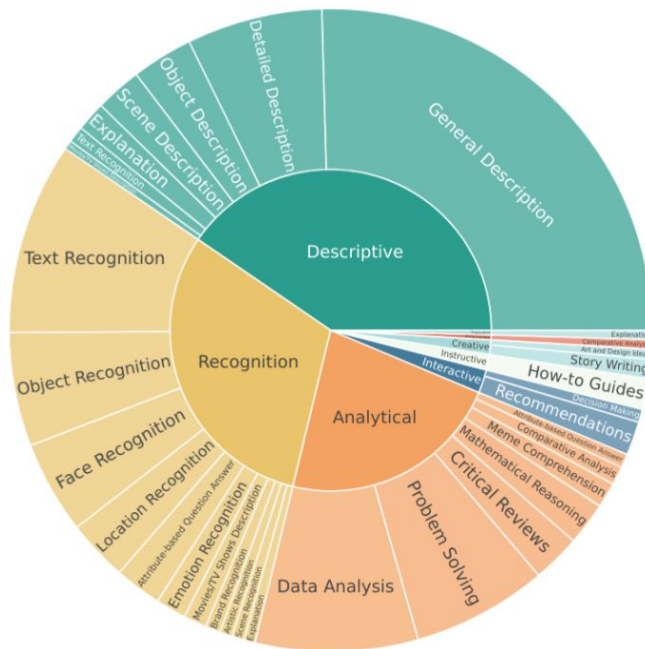


Figure 2: Question Category

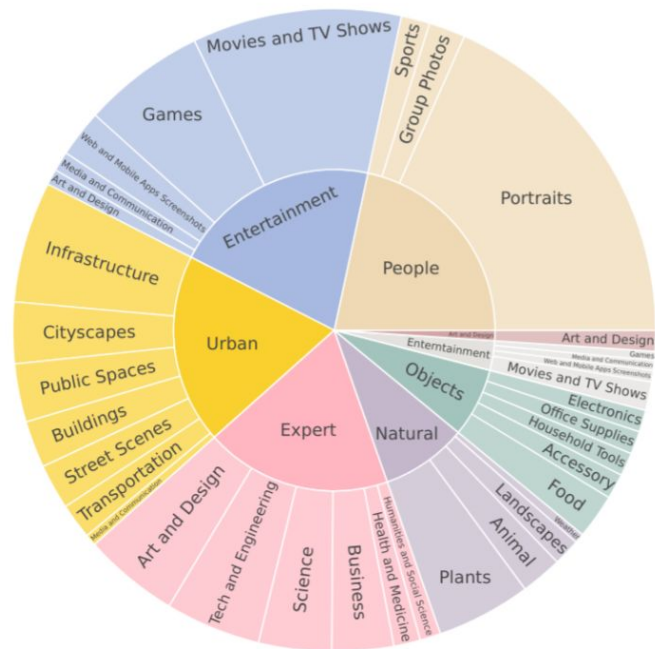


Figure 3: Image Domain

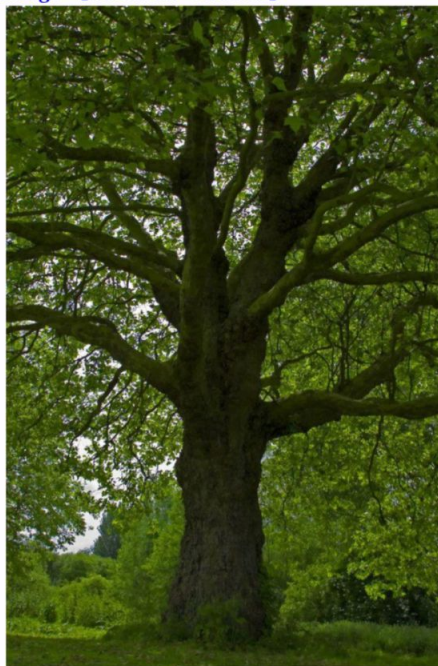
Benchmarks

Image [Entertainment-Movies/TV Shows]



[Descriptive-Movies/TV Shows] **Text Prompt:**
What are the two giraffe characters on this movie poster doing?

Image [Natural-Plants]



[Analytical-Problem Solving] **Text Prompt:**
How likely is it to snow after this picture was taken?
What would change with this type of tree before it's likely to snow?

Benchmarks

Image [Urban-Buildings]



[Recognition-Location] **Text Prompt:** where is this?

Image [Expert-Science]



[Analytical-Safety Procedures] **Text Prompt:** Can you tell me the potential risks and the unreasonable parts in the image?

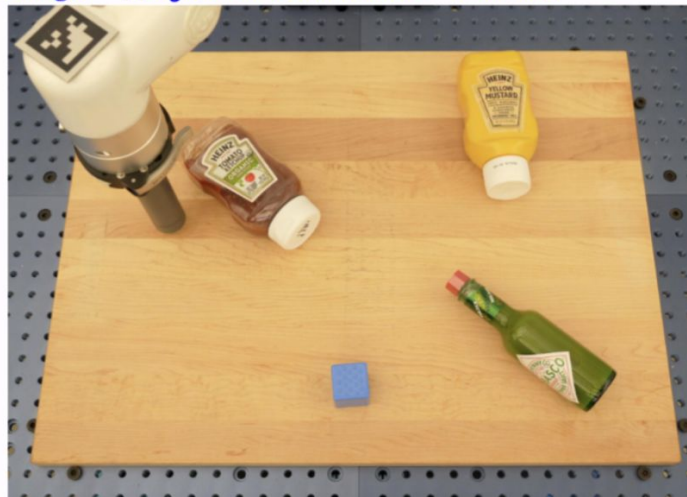
Benchmarks

Image [Natural-Landscapes]



[Recognition-Location] **Text Prompt:** where was this photo taken?

Image [Objects-Household Tools]



[Descriptive-Object Description] **Text Prompt:** describe the scene and objects

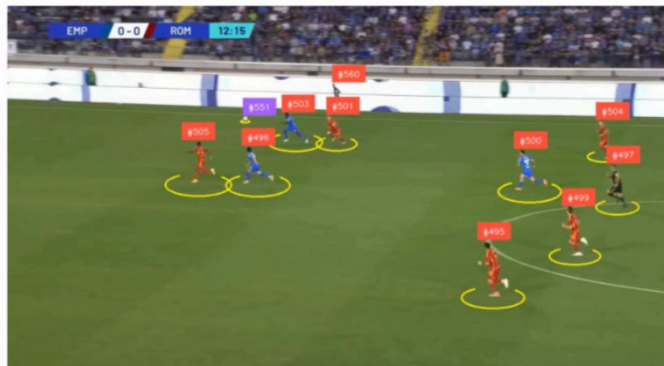
Benchmarks

Image [Entertainment-Web and Mobile Apps Screenshots]



[Interactive-Web Navigation] **Text Prompt:** I need to download flyer, you will be given screenshot from browser with elements marked with number. give next action to take on web page to download the flyers
give me response in below format example 1 action:[click,scroll,wait], box:1 format action:, box:

Image [Event-Sports]



[Descriptive-Scene Description] **Text Prompt:** this is a football match , every player has an identifier , describe every player action (example : player #501 is running)